

Recipe Management System - Backend Implementation

Project Overview

This backend application provides a robust API for exploring a collection of over 8,000 recipes. It is built with **Spring Boot** and uses an **H2 in-memory database** for seamless, zero-configuration setup by the reviewer.

Technical Logic & Implementation

1. Data Ingestion & Cleaning

- **Automated Loading:** On startup, a `DataIngestionService` reads the provided `recipes.json` file from the resources folder.
- **NaN Handling:** I implemented specific logic to identify "NaN" strings within numeric fields like `rating` and `prep_time`. These are converted to database NULL values to ensure mathematical integrity and prevent application crashes.
- **Database Population:** A total of **8,451 records** are successfully parsed and stored in the H2 `recipes` table.

2. API Design

- **Pagination & Sorting:** The `GET /api/recipes` endpoint uses `Spring Data Pageable` to return data in chunks, preventing memory overload. Results are automatically sorted by **rating (descending)**.
- **Advanced Search:** The `/search` endpoint allows optional filtering by **title**, **cuisine**, and **rating**. I used custom JPQL queries in the repository to handle these filters dynamically.
- **CORS Support:** Cross-Origin Resource Sharing is enabled to allow frontend integration.

Setup Instructions

1. **Prerequisites:** Java 17 or higher.
2. **Run Application:** Execute `./mvnw spring-boot:run` from the root directory.
3. **Backend Port:** The server runs on **port 8081**.

Testing the API

- **Get All:** GET `http://localhost:8081/api/recipes?page=1&limit=10`
- **Search:** GET `http://localhost:8081/api/recipes/search?title=pie&rating=4.0`
- **H2 Console:** Access at `http://localhost:8081/h2-console` with JDBC URL `jdbc:h2:mem:recipedb`.

OUTPUT:

- API END POINT 1: GET <http://localhost:8081/api/recipes?page=1&limit=10>

```
localhost:8081/api/recipes?page=1&limit=10

[{"content": [{"id": 2145, "cuisine": "Biscuit", "title": "Cheddar Keto Biscuits", "rating": 5.0, "prepTime": 20, "cookTime": 10, "totalTime": 40, "description": "Almond flour makes these keto cheddar cheese and garlic biscuits perfect for anyone following a diet or are trying to cut carbs out of their meals.", "nutrients": "{\"calories\":1244,\"carbohydrateContent\":4,\"cholesterolContent\":92,\"fiberContent\":1,\"proteinContent\":14,\"saturatedFatContent\":9,\"sodiumContent\":525,\"sugarContent\":1,\"fatContent\":20,\"unsaturatedFatContent\":0}", "serves": "8 servings"}, {"id": 2200, "cuisine": "Biscuit", "title": "Pepper Jack Cornbread Biscuits", "rating": 5.0, "prepTime": 30, "totalTime": 50, "description": "Buttery layered biscuits that have a slight taste of cornbread and a subtle spice of pepper jack cheese. These are perfect to go along with a soup or stew.", "nutrients": "{\"calories\":138,\"carbohydrateContent\":24,\"cholesterolContent\":21,\"fiberContent\":1,\"proteinContent\":3,\"saturatedFatContent\":5,\"sodiumContent\":411,\"sugarContent\":0,\"fatContent\":9,\"unsaturatedFatContent\":0}", "serves": "14 servings"}, {"id": 2284, "cuisine": "Monkey Bread", "title": "Double Chocolate Banana Monkey Bread", "rating": 5.0, "prepTime": 30, "cookTime": 35, "totalTime": 50, "description": "Double chocolate banana monkey bread is flaky layers of spiced, caramelized, double chocolate goodness that is a sweet, gooey delicacy.", "nutrients": "{\"calories\":395,\"carbohydrateContent\":60,\"cholesterolContent\":22,\"fiberContent\":2,\"proteinContent\":3,\"saturatedFatContent\":9,\"sodiumContent\":569,\"sugarContent\":41,\"fatContent\":17,\"unsaturatedFatContent\":0}", "serves": "12 servings"}, {"id": 2281, "cuisine": "Monkey Bread", "title": "Cliffy's Monkey Bread", "rating": 5.0, "prepTime": 35, "cookTime": 35, "totalTime": 70, "description": "A delicious monkey bread recipe that is perfect for a quick and easy snack or dessert. It's made with a mix of cinnamon rolls and topped with a delicious glaze.", "nutrients": "{\"calories\":389,\"carbohydrateContent\":48,\"cholesterolContent\":78,\"fiberContent\":3,\"proteinContent\":5,\"saturatedFatContent\":18,\"sodiumContent\":254,\"sugarContent\":20,\"fatContent\":21,\"unsaturatedFatContent\":0}", "serves": "8 servings"}], [{"id": 64, "cuisine": "Southern Desserts", "title": "Sweet Potato Pie", "rating": 4.8, "prepTime": 15, "cookTime": 100, "totalTime": 115, "description": "Shared from a Southern recipe, this homemade sweet potato pie is easy to make with boiled sweet potato. Try it, it may just be the best you've ever tasted!", "nutrients": "{\"calories\":389,\"carbohydrateContent\":48,\"cholesterolContent\":78,\"fiberContent\":3,\"proteinContent\":5,\"saturatedFatContent\":18,\"sodiumContent\":254,\"sugarContent\":20,\"fatContent\":21,\"unsaturatedFatContent\":0}", "serves": "8 servings"}, {"id": 76, "cuisine": "Southern Desserts", "title": "Grandma's Iron Skillet Apple Pie", "rating": 4.9, "prepTime": 15, "cookTime": 40, "totalTime": 75, "description": "This cast iron apple pie is a modern version of an old-time Southern favorite that uses premade pie crusts to make this three-layer apple pie.", "nutrients": "{\"calories\":734,\"carbohydrateContent\":100,\"cholesterolContent\":49,\"fiberContent\":2,\"proteinContent\":1,\"saturatedFatContent\":17,\"sodiumContent\":395,\"sugarContent\":70,\"fatContent\":34,\"unsaturatedFatContent\":0}", "serves": "8 servings"}, {"id": 83, "cuisine": "Southern Desserts", "title": "Butterwalk Chess Pie", "rating": 4.0, "prepTime": 30, "cookTime": 40, "totalTime": 70, "description": "A delicious chess pie recipe that is perfect for a quick and easy snack or dessert. It's made with a mix of cinnamon rolls and topped with a delicious glaze.", "nutrients": "{\"calories\":389,\"carbohydrateContent\":48,\"cholesterolContent\":78,\"fiberContent\":3,\"proteinContent\":5,\"saturatedFatContent\":18,\"sodiumContent\":254,\"sugarContent\":20,\"fatContent\":21,\"unsaturatedFatContent\":0}", "serves": "8 servings"}]
```

- API END POINT 2 : GET <http://localhost:8081/api/recipes/search?title=pie&rating=4.0>

```
localhost:8081/api/recipes/search?title=pie&rating=4.0

[{"content": [{"id": 1, "cuisine": "Southern Recipes", "title": "Sweet Potato Pie", "rating": 4.8, "prepTime": 15, "cookTime": 100, "totalTime": 115, "description": "Shared from a Southern recipe, this homemade sweet potato pie is easy to make with boiled sweet potato. Try it, it may just be the best you've ever tasted!", "nutrients": "{\"calories\":389,\"carbohydrateContent\":48,\"cholesterolContent\":78,\"fiberContent\":3,\"proteinContent\":5,\"saturatedFatContent\":18,\"sodiumContent\":254,\"sugarContent\":20,\"fatContent\":21,\"unsaturatedFatContent\":0}", "serves": "8 servings"}, {"id": 64, "cuisine": "Southern Desserts", "title": "Sweet Potato Pie", "rating": 4.8, "prepTime": 15, "cookTime": 100, "totalTime": 115, "description": "Shared from a Southern recipe, this homemade sweet potato pie is easy to make with boiled sweet potato. Try it, it may just be the best you've ever tasted!", "nutrients": "{\"calories\":389,\"carbohydrateContent\":48,\"cholesterolContent\":78,\"fiberContent\":3,\"proteinContent\":5,\"saturatedFatContent\":18,\"sodiumContent\":254,\"sugarContent\":20,\"fatContent\":21,\"unsaturatedFatContent\":0}", "serves": "8 servings"}, {"id": 76, "cuisine": "Southern Desserts", "title": "Grandma's Iron Skillet Apple Pie", "rating": 4.9, "prepTime": 15, "cookTime": 40, "totalTime": 75, "description": "This cast iron apple pie is a modern version of an old-time Southern favorite that uses premade pie crusts to make this three-layer apple pie.", "nutrients": "{\"calories\":734,\"carbohydrateContent\":100,\"cholesterolContent\":49,\"fiberContent\":2,\"proteinContent\":1,\"saturatedFatContent\":17,\"sodiumContent\":395,\"sugarContent\":70,\"fatContent\":34,\"unsaturatedFatContent\":0}", "serves": "8 servings"}, {"id": 83, "cuisine": "Southern Desserts", "title": "Butterwalk Chess Pie", "rating": 4.0, "prepTime": 30, "cookTime": 40, "totalTime": 70, "description": "A delicious chess pie recipe that is perfect for a quick and easy snack or dessert. It's made with a mix of cinnamon rolls and topped with a delicious glaze.", "nutrients": "{\"calories\":389,\"carbohydrateContent\":48,\"cholesterolContent\":78,\"fiberContent\":3,\"proteinContent\":5,\"saturatedFatContent\":18,\"sodiumContent\":254,\"sugarContent\":20,\"fatContent\":21,\"unsaturatedFatContent\":0}", "serves": "8 servings"}]
```

localhost:8081/h2-console/login.do?jsessionId=77b250cd0bad69db2b785e291a491583

Auto commit: ☒ Max rows: 1000 Auto complete: Off Auto select: On

jdbc:h2:mem:recipedb

- RECIPES
- INFORMATION_SCHEMA
- Users
- H2 2.2.224 (2023-09-17)

Run Run Selected Auto complete Clear SQL statement:

SELECT COUNT(*) FROM RECIPES;

SELECT COUNT(*) FROM RECIPES;
COUNT(*)
8451
(1 row, 2 ms)

localhost:8081/h2-console/login.do?jsessionId=42fac2f3a886cb2eec0ca4c38271a61e

Gmail YouTube Maps Merge k Sorted List...

English Preferences Tools Help

Login

Saved Settings: Generic H2 (Embedded)

Setting Name: Generic H2 (Embedded) Save Remove

Driver Class: org.h2.Driver

JDBC URL: jdbc:h2:mem:recipedb

User Name: sa

Password:

Connect Test Connection